

Semi-Humanoid Robotic Platform

Bill of Materials & Device-Level Cost Breakdown

1. Design, Sourcing & Engineering Notes

- **Arms:** 6-DoF each (shoulder pitch/roll/yaw, elbow pitch, wrist pitch/roll) plus a 1-DoF gripper. The four large shoulder/elbow joints use CubeMars AK70-10 quasi-direct-drive (QDD) actuators, whose peak torque is sized for the arm's target payload at its working link lengths; the two wrist joints and the gripper use Feetech serial bus servos, which deliver adequate torque for end-effector-frame loads once the payload is already carried by the larger proximal joints – at a fraction of the cost of premium smart servos.
- **Head:** 2-DoF pan/tilt on Feetech STS3215 bus servos (sized for a light sensor-head payload), carrying an Intel RealSense depth/RGB camera, matched to the manipulation/perception pipeline.
- **Base:** differential drive on two 100 W DC gearmotors sized for the base's mass and target travel speed, standard wheels, a 2D LiDAR for SLAM/navigation, and its own dedicated controller.
- **Torso:** houses the waist tilt joint (a second AK70-10, torque-matched to lift the full upper-body mass), the main structural frame, the Jetson Orin NX compute module, and – per the current design – the main battery pack and all power-distribution, charging and safety-cutoff hardware.
- **Telescopic column:** a separate, optional lift stage between the base and torso, thrust-rated with margin above the mass it lifts. Costed independently so it can be added to or dropped from the build without disturbing the core BOM.

Core platform total (Base + Torso + Head + 2 Arms)	EGP 432,915
Telescopic column add-on (optional)	+EGP 32,920 (+7.6%)
Grand total with telescopic column	EGP 465,835

2. Table 1 — Consolidated Bill of Materials

Component	Specification	Device	Qty	Unit Cost	Total Cost
Actuation					
DC Gearmotor with Encoder	24 V, 100 W, ~30:1 planetary reduction, quadrature encoder. Torque/speed sized for a 35–45 kg base at typical indoor travel speeds (0.6–1.0 m/s).	Base	2	EGP 7,405	EGP 14,810
Quasi-Direct-Drive (QDD) Actuator	CubeMars AK70-10, 24.8 N·m peak, 10:1 planetary, integrated FOC driver + 14-bit encoder – distributed large-joint actuation (waist, shoulder pitch/roll/yaw, and elbow pitch).	Torso, Arm (x2)	9	EGP 22,155	EGP 199,395
Serial Bus Servo (STS3215)	Feetech STS3215, 12 V, 2.9 N·m stall – sized for head pan/tilt axes and arm wrist pitch/roll axes.	Head, Arm (x2)	6	EGP 1,480	EGP 8,880
Serial Bus Servo (STS3032)	Feetech STS3032, compact bus servo – drives the parallel-jaw gripper.	Arm (x2)	2	EGP 1,075	EGP 2,150
Sensing & Perception					
2D LiDAR	Slamtec RPLiDAR A1M8, 12 m range, 360° scan – range and scan rate matched to room-scale SLAM/navigation.	Base	1	EGP 6,665	EGP 6,665
Ultrasonic Proximity Sensor	HC-SR04 (4 units) – short-range redundancy for the LiDAR's blind zone directly around the base.	Base	4	EGP 205	EGP 820
Torso IMU	9-DOF IMU breakout (BNO085-class) – orientation/balance feedback at a loop rate matched to the waist controller.	Torso	1	EGP 930	EGP 930
RGB-D Perception Camera	Intel RealSense – depth and RGB sensing, matched to the manipulation/perception pipeline's needs.	Head	1	EGP 55,000	EGP 55,000
Microphone Array	ReSpeaker 2-Mic Pi HAT – far-field voice pickup for human-robot interaction.	Head	1	EGP 1,685	EGP 1,685
Speaker	2 W speaker with I2S/analog amplifier – robot voice/audio feedback.	Head	1	EGP 620	EGP 620
Compute & Control					
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC – distributed embedded controllers for per-subsystem local control loops (drivetrain, waist/IMU, head, and arms).	Base, Torso, Head, Arm (x2)	5	EGP 465	EGP 2,325
Custom Base Controller PCB	Populated PCB integrating the dual-motor H-bridge power stage, quadrature-encoder signal conditioning, and an ESP32 carrier header; fabricated and assembled to spec.	Base	1	EGP 3,030	EGP 3,030

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Component	Specification	Device	Qty	Unit Cost	Total Cost
Main Compute Module	NVIDIA Jetson Orin NX 16 GB (module + carrier-board developer kit) – compute headroom matched to the perception, navigation, manipulation and VLA workload targeted by the project.	Torso	1	EGP 100,000	EGP 100,000
Compute Cooling Kit	Active fan + heatsink, sized for sustained Jetson load inside an enclosed torso bay.	Torso	1	EGP 1,815	EGP 1,815
Custom Power Distribution & Waist-Controller PCB	Populated PCB integrating fused battery-rail distribution, three 24 V→12 V/5 V synchronous buck regulators, a 3.3 V CAN transceiver wired to the ESP32's native TWAI controller (waist QDD bus), and status/interlock circuitry.	Torso	1	EGP 4,715	EGP 4,715
Custom Head Interface PCB	Populated PCB integrating a half-duplex TTL bus-servo header (Feetech pan/tilt chain), an audio-amplifier stage, and an ESP32 carrier header.	Head	1	EGP 1,885	EGP 1,885
Custom Arm Joint Interface PCB	Populated PCB per arm integrating a 3.3 V CAN transceiver on the ESP32's native TWAI bus (QDD actuator chain) and the half-duplex TTL bus-servo header (wrist + gripper chain).	Arm (x2)	2	EGP 2,155	EGP 4,310
Power & Electrical					
Main Battery Pack	LiFePO4, 24 V, 20 Ah (~480 Wh), integrated BMS – capacity sized for a multi-hour duty cycle across drivetrain, compute and actuators.	Torso	1	EGP 13,935	EGP 13,935
Battery Smart Charger	24 V LiFePO4-profile smart charger, 5 A.	Torso	1	EGP 2,330	EGP 2,330
Main Power Contactor / Relay	24 V coil, 40 A continuous – rated above the platform's worst-case simultaneous current draw; master power cut-off.	Torso	1	EGP 1,815	EGP 1,815
Structural & Mechanical					
Drive Wheel	200 mm rubber-tread wheel, motor-shaft hub, standard (non-mecanum) – diameter chosen to clear small indoor thresholds and cabling.	Base	2	EGP 1,970	EGP 3,940
Swivel Caster Wheel	100 mm load-rated caster – rear support point for a stable three/four-point base.	Base	2	EGP 930	EGP 1,860
CORE PLATFORM TOTAL (Base + Torso + Head + Arms)					EGP 432,915

3. Table 2 — Device-Level Cost Breakdown

3.1 Base (Mobile Platform)

Units on platform: x1

Component	Specification	Qty	Unit Cost	Line Cost	Category
DC Gearmotor with Encoder	24 V, 100 W, ~30:1 planetary reduction, quadrature encoder. Torque/speed sized for a 35–45 kg base at typical indoor travel speeds (0.6–1.0 m/s).	2	EGP 7,405	EGP 14,810	Actuation
Drive Wheel	200 mm rubber-tread wheel, motor-shaft hub, standard (non-mecanum) – diameter chosen to clear small indoor thresholds and cabling.	2	EGP 1,970	EGP 3,940	Structural & Mechanical
Swivel Caster Wheel	100 mm load-rated caster – rear support point for a stable three/four-point base.	2	EGP 930	EGP 1,860	Structural & Mechanical
2D LiDAR	Slamtec RPLiDAR A1M8, 12 m range, 360° scan – range and scan rate matched to room-scale SLAM/navigation.	1	EGP 6,665	EGP 6,665	Sensing & Perception
Ultrasonic Proximity Sensor	HC-SR04 (4 units) – short-range redundancy for the LiDAR's blind zone directly around the base.	4	EGP 205	EGP 820	Sensing & Perception
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC – closes the drive-motor PID loop and reads both wheel encoders at the update rate the base controller needs.	1	EGP 465	EGP 465	Compute & Control
Custom Base Controller PCB	Populated PCB integrating the dual-motor H-bridge power stage, quadrature-encoder signal conditioning, and an ESP32 carrier header; fabricated and assembled to spec.	1	EGP 3,030	EGP 3,030	Compute & Control
Device total cost				EGP 31,590	7.3% of the core platform cost (EGP 432,915)

3.2 Torso (incl. Waist Joint, Compute & Power)

Component	Specification	Qty	Unit Cost	Line Cost	Category
Quasi-Direct-Drive (QDD) Actuator	CubeMars AK70-10 – 24.8 N·m peak, 10:1 planetary, integrated FOC driver + 14-bit encoder – waist tilt joint; peak torque sized to tilt the full torso+arm+head mass through its working range.	1	EGP 22,155	EGP 22,155	Actuation
Torso IMU	9-DOF IMU breakout (BNO085-class) – orientation/balance feedback at a loop rate matched to the waist controller.	1	EGP 930	EGP 930	Sensing & Perception
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC – waist-joint control, torso IMU acquisition, power/safety interlocks.	1	EGP 465	EGP 465	Compute & Control
Main Compute Module	NVIDIA Jetson Orin NX 16 GB (module + carrier-board developer kit) – compute headroom matched to the perception, navigation, manipulation and VLA workload targeted by the project.	1	EGP 100,000	EGP 100,000	Compute & Control
Compute Cooling Kit	Active fan + heatsink, sized for sustained Jetson load inside an enclosed torso bay.	1	EGP 1,815	EGP 1,815	Compute & Control
Main Battery Pack	LiFePO ₄ , 24 V, 20 Ah (~480 Wh), integrated BMS – capacity sized for a multi-hour duty cycle across drivetrain, compute and actuators.	1	EGP 13,935	EGP 13,935	Power & Electrical
Battery Smart Charger	24 V LiFePO ₄ -profile smart charger, 5 A.	1	EGP 2,330	EGP 2,330	Power & Electrical
Main Power Contactor / Relay	24 V coil, 40 A continuous – rated above the platform's worst-case simultaneous current draw; master power cut-off.	1	EGP 1,815	EGP 1,815	Power & Electrical
Custom Power Distribution & Waist-Controller PCB	Populated PCB integrating fused battery-rail distribution, three 24 V→12 V/5 V synchronous buck regulators, a 3.3 V CAN transceiver wired to the ESP32's native TWAI controller (waist QDD bus), and status/interlock circuitry.	1	EGP 4,715	EGP 4,715	Compute & Control
Device total cost				EGP 148,160	34.2% of the core platform cost (EGP 432,915)

3.3 Head (Perception Head)

Component	Specification	Qty	Unit Cost	Line Cost	Category
Serial Bus Servo – Pan	Feetech STS3215, 12 V, 2.9 N·m stall – torque sized for a light sensor-head payload (camera + mic + speaker) at low pan speed.	1	EGP 1,480	EGP 1,480	Actuation
Serial Bus Servo – Tilt	Feetech STS3215, 12 V – head pitch axis, same sizing rationale as the pan axis.	1	EGP 1,480	EGP 1,480	Actuation
RGB-D Perception Camera	Intel RealSense – depth and RGB sensing, matched to the manipulation/perception pipeline's needs.	1	EGP 55,000	EGP 55,000	Sensing & Perception
Microphone Array	ReSpeaker 2-Mic Pi HAT – far-field voice pickup for human-robot interaction.	1	EGP 1,685	EGP 1,685	Sensing & Perception
Speaker	2 W speaker with I2S/analog amplifier – robot voice/audio feedback.	1	EGP 620	EGP 620	Sensing & Perception
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC – drives the pan/tilt servos and manages microphone/speaker I2S audio.	1	EGP 465	EGP 465	Compute & Control
Custom Head Interface PCB	Populated PCB integrating a half-duplex TTL bus-servo header (Feetech pan/tilt chain), an audio-amplifier stage, and an ESP32 carrier header.	1	EGP 1,885	EGP 1,885	Compute & Control
Device total cost				EGP 62,615	14.5% of the core platform cost (EGP 432,915)

3.4 Arm (6-DoF Manipulator)

Component	Specification	Qty	Unit Cost	Line Cost	Category
Quasi-Direct-Drive (QDD) Actuator	CubeMars AK70-10, 24.8 N·m peak – shoulder pitch/roll/yaw and elbow pitch (4 per arm); peak torque sized for the arm's target payload at its working link lengths.	8	EGP 22,155	EGP 177,240	Actuation
Serial Bus Servo – Wrist	Feetech STS3215, wrist pitch and roll (2 per arm) – adequate once payload has already been carried by the larger proximal joints.	4	EGP 1,480	EGP 5,920	Actuation
Serial Bus Servo – Gripper	Feetech STS3032, compact bus servo – drives the parallel-jaw gripper (1 per arm).	2	EGP 1,075	EGP 2,150	Actuation
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC, 1 per arm – per-arm joint coordinator.	2	EGP 465	EGP 930	Compute & Control
Custom Arm Joint Interface PCB	Populated PCB per arm integrating a 3.3 V CAN transceiver on the ESP32's native TWAI bus (QDD actuator chain) and the half-duplex TTL bus-servo header (wrist + gripper chain).	2	EGP 2,155	EGP 4,310	Compute & Control
Device total cost				EGP 190,550	44.0% of the core platform cost (EGP 432,915)

3.5 Telescopic Column (Optional Module)

Units on platform: x1, OPTIONAL

Component	Specification	Qty	Unit Cost	Line Cost	Category
Heavy-Duty Linear Actuator	24 V electric linear actuator, 300 mm stroke, 1500 N thrust – rated with margin above the torso+arms+head weight it lifts.	1	EGP 17,510	EGP 17,510	Actuation
Linear Guide Rail + Carriage	Guides the telescoping column and takes side-loads off the actuator rod.	1	EGP 6,215	EGP 6,215	Structural & Mechanical
Telescoping Column Tube Set	Nested aluminum tube sections with bearings/bushings.	1	EGP 7,250	EGP 7,250	Structural & Mechanical
Dedicated Microcontroller	ESP32-WROOM-32 DevKitC – independent telescopic-column position control.	1	EGP 465	EGP 465	Compute & Control
Custom Column Driver PCB	Populated PCB integrating an H-bridge power stage for the linear actuator and an ESP32 carrier header.	1	EGP 1,480	EGP 1,480	Compute & Control
Device total cost				EGP 32,920	adds 7.6% to the core platform cost

3.6 Cost Share Summary

Device	Total Cost (EGP)	% of Core Platform Cost
Arm (x2, combined)	EGP 190,550	44.0%
Torso	EGP 148,160	34.2%
Base	EGP 31,590	7.3%
Head	EGP 62,615	14.5%
CORE PLATFORM TOTAL	EGP 432,915	100.0%
Telescopic column (optional add-on)	+EGP 32,920	+7.6% increase
GRAND TOTAL incl. telescopic column	EGP 465,835	—